

SpatialLens Assist: Motion-Aware Hazard Detection for Campus Navigation from Low-FPS Egocentric Video

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Abstract

An object detector tells a blind pedestrian *what* is nearby, but not whether it is closing, crossing, or simply present. We ask whether a small, interpretable set of motion cues—bounding-box growth, centroid displacement, optical flow, and ego-motion compensation—can recover useful hazard labels from low-fps handheld campus video, without training a learned trajectory model. We present **SpatialLens Assist**, an end-to-end CPU pipeline that converts short (~ 10 s) walkway clips into per-track motion-aware alerts over six classes (**approaching**, **crossing_LTR**, **crossingRTL**, **moving_away**, **static**, **uncertain**). The system pairs YOLOv8n with an IoU+centroid tracker, ego-motion-compensated Farneback flow and frame differencing, per-track motion aggregates, and a transparent rule cascade that emits a natural-language evidence string and treats **uncertain** as a first-class abstention rather than a guess. On 11 controlled walkway videos (61 manually labelled tracks), the system reaches **82.5%** overall accuracy on 57 evaluated track-label pairs and **93.3%** on the 30 tracks with at least three frames, with approaching precision/recall **0.95/0.88**—the safety-asymmetric regime an alert layer needs. Ablations show the largest gain comes from tracker continuity (+6.2 pts from appearance-gated re-identification, +3.5 pts each from two short-track salvage rules), and ByteTrack is a negative result on this sparse 2 fps regime (32.8%). The dominant failure mode is conservative abstention from tracker fragmentation, not dangerous mislabelling: *no approaching* $\leftrightarrow$ *moving_away* confusion occurs.

1 Introduction

A bicycle parked against a wall and a bicycle bearing down on the camera are both detected by a standard detector as “bicycle, 0.86.” For a sighted user the wider visual context carries the intent; for a blind or low-vision pedestrian relying on an assistive device, that wider context is exactly what is missing. The safety-critical signal is not the class label but the *motion*: whether the agent is closing, crossing the user’s path, or merely present.

Research question. SpatialLens Assist asks whether lightweight, interpretable motion cues are sufficient to classify pedestrian-scale hazards in low-fps egocentric campus video. In contrast to learned trajectory classifiers that require larger labelled datasets, we use a transparent rule cascade over ego-motion-compensated optical flow, centroid displacement, and bounding-box growth. Rather than training a learned trajectory classifier on a small dataset, SpatialLens evaluates whether a transparent rule cascade can achieve high accuracy on decidable tracks while abstaining when temporal evidence is insufficient.

Approach in one paragraph. We frame motion-aware hazard detection as a *per-track classification* problem: given a ~ 10 s phone clip sampled at 2 fps, decide for each tracked object whether it is approaching, crossing left-to-right, crossing right-to-left, moving away, static, or—when temporal evidence is too sparse—explicitly **uncertain**. Three design properties drive the system: (i) **explainability**—every prediction carries a textual reason the alert layer can quote; (ii) **calibrated abstention**—short tracks should not bluff, so **uncertain** is a first-class output with a sub-reason (**short_track**, **low_signal**, ...); and (iii) **reproducibility**—the pipeline runs in ~ 4 min on a GPU-less M-series Mac, and the 61-track labelled set is committed.

Summary of findings. On 11 controlled campus walkway videos with 61 manually-labelled tracks, the system reaches 82.5% accuracy on 57 matched tracks and 93.3% on the 30 tracks with at least three frames; the safety-critical **approaching** class reaches 0.95 / 0.88 precision/recall; *no approaching* $\leftrightarrow$ *moving_away* confusion is observed. Ablations attribute the dominant accuracy gain to track continuity rather than classifier complexity: appearance-gated fragment re-identification contributes 6.2 pts and the two 2-frame salvage rules contribute 3.5 pts each, while ByteTrack on this 2 fps offline footage drops to 32.8%—a useful negative result.

Contributions.

- C1.** A complete offline egocentric video pipeline for motion-aware campus hazard alerts, evaluable end-to-end on CPU in ~ 4 minutes.
- C2.** Ego-motion-compensated motion features (Farneback flow with per-frame median subtraction; ECC-aligned frame differencing) appropriate to handheld low-fps footage.
- C3.** An interpretable hazard rule cascade with natural-language evidence strings and conservative abstention, including a trajectory-reversal heuristic for “approached-then-turned-away” tracks.
- C4.** A manually-labelled 61-track evaluation set across 11 campus videos, with two reporting denominators (raw and per-physical-object consolidation) and a fully scripted ablation suite.
- C5.** Empirical evidence that on this regime tracking continuity dominates classifier sophistication: appearance-gated re-ID and short-track salvage are the largest gains, and ByteTrack is poorly matched to sparse offline footage.

2 Related Work

Object detection and tracking for assistive navigation. YOLOv8 [1] is the fixed COCO-class detector front-end used here; trackers such as SORT [3] and ByteTrack [2] are strong online baselines but assume near-continuous per-frame detections from a Kalman model. On our 2 fps offline footage ByteTrack (via **supervision** [4]) drops 51% of detections under defaults—a known limitation of motion-continuity assumptions in sparse sampling regimes [2]. We therefore implement a transparent per-class IoU+centroid matcher with an explicit gap buffer and a separate appearance-gated fragment re-ID pass tailored to the sparse-sampling regime. Assistive vision tools such as Seeing AI [7] and academic prototypes [8] expose object *identity* but generally not *motion intent* relative to the user; their alerts read “person” or “bicycle,” not “bike crossing right.”

Optical flow and ego-motion for motion reasoning. Classical Farneback flow [9] and ECC alignment [10] are well suited to the CPU-only constraint of an offline assistive prototype. We use both, and explicitly subtract the per-frame median flow vector to compensate for the camera pan that dominates handheld footage—without this step, every static object exhibits large apparent motion. The same idea has been used in classical SLAM and recent learned ego-motion estimation [11]; we deliberately stay in the classical regime to keep the pipeline GPU-free and interpretable.

Explainable rule-based vs. learned trajectory classifiers. Modern trajectory forecasters such as Trajectron++ [5] and Social-STGCNN [6] predict *future* positions over 4–8 s from high-fps tracks with map context and require thousands of labelled trajectories—more data than this project collects, and an output format (continuous future positions) mismatched to a spoken hazard alert. We solve the strictly easier problem of classifying the *past* motion of a single track into six safety-relevant categories from ~ 10 sampled frames, and we choose to output an alert with an evidence string rather than a future-position distribution. This is a deliberate scope decision driven by (i) the no-training-data setting, (ii) the explainability requirement of an assistive interface, and (iii) the conservative abstention requirement when temporal evidence is sparse.

Why low-fps egocentric footage is challenging. At 2 fps a fast-crossing object can move >300 px between samples; Kalman trackers lose the gate, IoU trackers cannot match the new detection to the prior, and naive bbox-growth features confound camera pan with object motion. These three failure modes motivate the three corresponding design choices in Section 3: an explicit gap buffer, a conservative post-tracker re-ID pass, and per-frame ego-motion subtraction.

3 Method

The pipeline (Figure 1) has six stages. We summarise each, focusing on the four non-obvious design decisions.

3.1 Frames, detection, and base tracking

Videos are sampled at 2 fps and resized to 960 px wide. YOLOv8n emits per-frame detections at confidence ≥ 0.35 (lower thresholds produced many 1–2 frame flicker tracks). Detections feed a **per-class IoU+centroid tracker**: for each detection d and active track t of the same class, we score $s_{dt} = 0.7 \cdot \text{IoU}(d, t) + 0.3 \cdot (1 - \hat{d}_{dt})$, where $\hat{d}_{dt}$ is centroid distance normalised by the image diagonal. Greedy assignment in score order, with a configurable `max_frame_gap` buffer so a track survives 1–2 missed detections.

3.2 Motion features with ego-motion compensation

Handheld footage translates the whole image during a pan, so every static object shows large apparent flow. We compensate two ways. **Flow:** Farneback dense flow between frames, minus the *per-frame median* flow vector (background-dominated), giving per-bbox (flow_dx, flow_dy, flow_mag) relative to the scene. **Frame diff:** we align the previous frame to the current with `cv2.findTransformECC` (translation) before differencing, removing most pan-induced false-positive mask area.

Per-track aggregates include the start-to-end bbox-growth ratio $r = (A_{\text{end}} - A_{\text{start}})/A_{\text{start}}$, total Δc_x , mean flow magnitude inside the bbox, and two sub-track signals introduced for trajectory-reversal detection: **second-half growth ratio** $r_{1/2}$ (growth from the middle frame’s area to the last frame’s area) and **peak-area ratio** $\rho = \max_t A_t/A_{\text{end}}$.

3.3 Conservative re-identification of fragments

At 2 fps a fast-crossing object can move 350 px between samples, so the IoU+centroid tracker fragments the same physical object into several short tracks. We add a *post-tracker* cleanup pass with two stages:

1. **Same-frame NMS.** Drop overlapping same-class detections within a frame at $\text{IoU} \geq 0.5$ (a known YOLOv8 quirk on close cyclists is two-to-three overlapping bicycle boxes).
2. **Appearance-gated stitching.** For each pair of tracks (A, B) with B starting after A ends, merge $B \rightarrow A$ only if *all* of: (a) same class, (b) *both tracks are exactly one frame long*, (c) gap ≤ 2 frames, (d) centroid distance ≤ 0.45 diag, (e) bbox-area ratio ≤ 5 , and (f) HSV (H, S) histogram correlation ≥ 0.45 . Constraint (b) protects every multi-frame track from being absorbed by a fragment—an aggressive variant tried early in development broke seven previously-correct tracks (Section 5.4).

The cleanup is a separate module (`src/track_reid.py`) so the tracker stays auditable and ablations remain straightforward.

3.4 Rule-cascade hazard classifier

The classifier resolves the label by a priority cascade, each branch carrying its own evidence template (r : bbox-growth ratio; ρ : peak/end area ratio; $r_{1/2}$: second-half

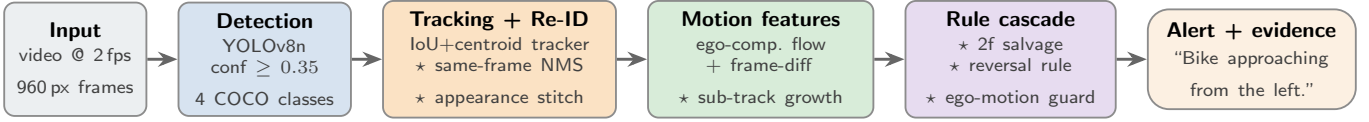


Figure 1: End-to-end SpatialLens pipeline. The detector is off-the-shelf; ★ marks the four contributions evaluated in Section 5 (same-frame NMS, appearance-gated re-ID, sub-track growth features, and the rule-cascade salvage/guard rules). Every prediction carries a natural-language *evidence string* that the alert layer quotes, and **uncertain** is a first-class output with a sub-reason rather than a fall-through default.

growth): (1) **Short-track salvage** ($n=2$): strong lateral $|dx_norm| > 0.30 \rightarrow$ crossing, else strong approach ($r > 0.7$, downward flow) $\rightarrow$ approaching, else **uncertain**. (2) **Static** if $n \geq 3$ and no motion. (3) **Moving-away** if $r < -0.15$, or the *soft* variant ($-0.15 < r < -0.03$ with sub-strong lateral motion read as ego-motion drift). (4) **Trajectory reversal** ($n \geq 6$, $r > 0.3$, $\rho > 1.5$, $r_{1/2} < -0.20$) $\rightarrow$ moving_away for the “approached then turned away” case. (5) **Strong crossing** ($|dx_norm| > 0.50$) overrides approach. (6) **Approaching** if $approach_score \geq 0.55$ and ($r > 0.15$ or the slow-approach allowance $r > 0$ with $score \geq 0.65$). (7) **Crossing** if $|dx_norm| > 0.08$, $r \leq 0.15$. (8) **Uncertain** fall-through, tagged with the closest-missed sub-reason (near_approaching, low_signal, ...) quoted in the evidence string.

The *approach score* is a fixed-weight blend $0.45 \cdot s_{growth} + 0.25 \cdot s_{center} + 0.20 \cdot s_{fd} + 0.10 \cdot s_{flow}$ of four normalised cues (bbox growth, centroid motion toward image center/lower half, frame-diff overlap, and flow magnitude).

3.5 Alert generation and explainability

Every prediction is paired with a templated *evidence string* quoting the specific feature values that triggered the cascade branch (e.g. “bicycle_1 classified as approaching because bbox grew by 11.42 and approach_score = 1.00; centroid moved toward image center/lower half; flow_mag = 17.26”). For abstentions the evidence string names the sub-reason: “classified as uncertain (short.track) because track is only 2 frames long (min = 3), so trajectory and bbox-growth cues are unreliable.” Because the application is assistive navigation, unsupported motion claims are undesirable: we therefore treat uncertainty as a safety-preserving abstention—tracks with fewer than three frames are not forced into a motion class.

Figure 2 visualises the mechanism on a representative approaching track: across four sampled frames of a 15-frame cyclist, the bounding box grows by 21.98 $\times$, the yellow centroid trail moves toward the image centre, and the approach score saturates at 1.00, yielding an unambiguous approaching call.

4 Dataset and Evaluation Protocol

4.1 Videos and tracks

We recorded 11 controlled ~ 10 s phone videos on Stanford walkways under daylight, sampling at 2 fps to 960 px-wide frames (Table 1). Each captures a single scripted hazard

mode (bike approaching, pedestrian crossing, etc.) so that label intent is unambiguous. Two additional natural-walk clips are used for qualitative demonstration only and are excluded from the quantitative tables.

4.2 Track-level labels and class distribution

Every detected, tracked object in the 11 controlled videos was labelled by hand into the six-class taxonomy described in the Introduction (Table 2). The six classes are deliberately motion-centric: approaching, moving_away, crossing_left_to_right, crossing_right_to_left, static, and uncertain. The class distribution is imbalanced—approaching and uncertain dominate, while static has only two examples (both short tracker fragments of parked bicycles). This imbalance is realistic for campus walkway footage but informs how the per-class numbers should be read (Sections 5 and 6).

4.3 Track-label matching and evaluated denominator

Predictions and labels are matched by (video_id, track_id) pairs. After post-tracker re-identification, 4 of the 61 labelled tracks were absorbed by NMS/re-ID into an existing labelled track (their detections moved into a different track_id), so 57 prediction-label pairs remain for the raw-denominator evaluation. We also publish a 51-track consolidated label set in which the labeller’s notes column marked 9 entries as “fragmentation” or “duplicate”; these collapse cleanly into one representative each. The raw 82.5% is the metric to compare against any system without re-ID; the consolidated 92.2% reads as accuracy *per unique physical object*. Both label sets are versioned in the repository.

4.4 Metrics

We report overall accuracy on the 57 evaluated tracks; the *decidable* accuracy on the 30 tracks with at least 3 frames (the regime in which the cascade has enough temporal evidence to classify motion); per-class precision/recall/F1 and macro-F1; the full 6×6 confusion matrix; and an ablation suite that toggles individual contributions one at a time. We use sklearn-compatible support counts, and we report two reporting denominators (raw and consolidated) rather than picking the flattering one. Macro-F1 is lower than accuracy because rare classes such as static and crossing RTL have small support; macro-F1 weights every class equally, while accuracy is dominated by approaching and uncertain tracks.

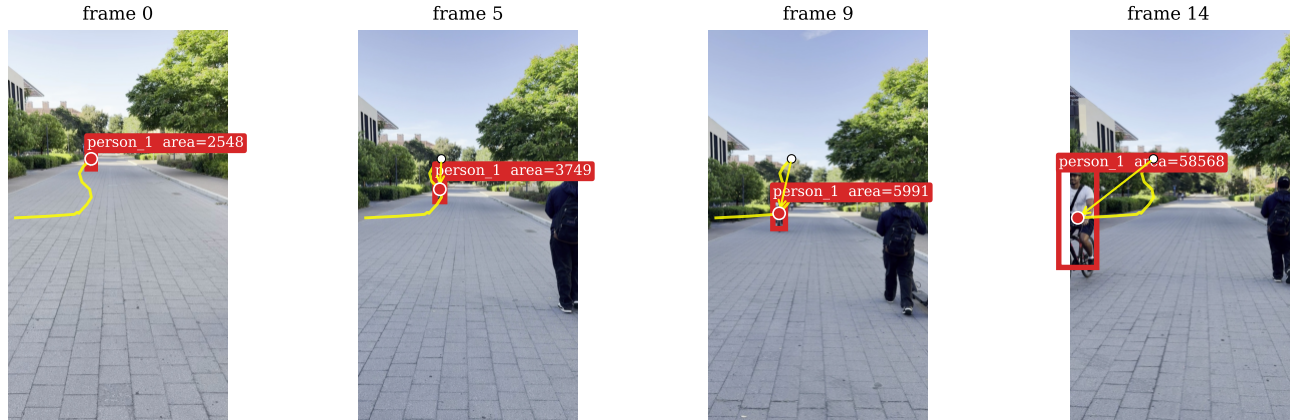

 Evidence — bbox growth $r = +21.98$, $dx_{norm} = -0.15$, flow mag = 14.5, approach score = 1.00 $\Rightarrow$ label = APPROACHING

Figure 2: Temporal motion evidence for an **approaching** cyclist (IMG_4972 / person_1, 15 frames). The bounding-box area grows monotonically across the strip ($A = 2548 \rightarrow 58568 \text{ px}^2$, $r = +21.98$); the yellow centroid trail moves toward the lower centre of the frame; and ego-motion-compensated flow remains consistent with object motion rather than camera pan. The evidence string is what the alert layer quotes verbatim, making the decision auditable.

Table 1: Dataset splits used in this paper. The 11 controlled campus videos provide all 61 manually-labelled tracks; 4 labels are absorbed by post-tracker re-ID, leaving 57 prediction-label pairs for the raw denominator and 30 tracks for the decidable subset.

Split	Videos	Labeled tracks	Used quantitatively?	Notes
Controlled campus videos	11	61	Yes	2 fps, 960 px wide, hand-held phone
Matched evaluated tracks	11	57	Yes	after track-label matching
Decidable subset ($n \geq 3$)	11	30	Yes	tracks with ≥ 3 frames
Uncontrolled qualitative	2	—	No	natural-walk clips, demo only

Table 2: Ground-truth class distribution. The set is imbalanced: **approaching** dominates; **static** and **crossing_RTL** have $n \leq 3$.

Class	n	%	Note
approaching	24	39.3%	primary safety-critical class
crossing_left_to_right	7	11.5%	limited support
crossing_right_to_left	3	4.9%	limited support
moving_away	9	14.8%	non-hazard motion
static	2	3.3%	only 2 examples $\Rightarrow$ high variance
uncertain	16	26.2%	deliberate abstention label
Total	61	100.0%	11 controlled videos

Table 3: Per-class precision, recall, F1, and support (raw 57-track denominator). Macro-F1 0.67 is dominated by the two rare classes (**crossing_RTL**, **static**) with $n \leq 3$.

Class	Precision	Recall	F1	n
approaching	0.95	0.88	0.91	24
crossing_left_to_right	1.00	0.50	0.67	4
crossing_right_to_left	1.00	0.67	0.80	3
moving_away	1.00	0.78	0.88	9
static	0.00	0.00	0.00	2
uncertain	0.62	1.00	0.77	15
Overall accuracy	82.5% on 57 evaluated tracks			
Decidable accuracy ($n \geq 3$)	93.3% on 30 tracks			
Macro F1	0.67			

5 Results

5.1 Main quantitative results

Table 3 reports the per-class metrics. The safety-critical **approaching** class reaches **0.95 / 0.88** precision/recall—an asymmetry an alert layer wants, since false alarms are recoverable but missed approachers are not. Every non-trivial directional class has precision 1.00, so when the cascade commits to a directional label it is essentially never wrong; what it does instead, when temporal evidence is thin, is abstain into **uncertain**. The gap between overall accuracy (82.5%) and decidable accuracy (93.3%) is explained primarily by short tracker fragments: when at least three frames are available the rule cascade has enough temporal evidence to classify motion reliably; when only one or two detections are linked, the system usually abstains.

Results are not single-video luck. 6 of the 11 videos are

classified perfectly, and per-video accuracy ranges from 1/3 (IMG_4977, the densest fast-crossing fragmentation) to 16/16; the worst absolute count is IMG_4982 (11/16 raw) where overlapping pedestrians and parked bicycles fragment into 1-frame tracks the re-ID gate is too conservative to merge.

5.2 Motion feature space

Figure 4 plots every labelled track in the (dx_{norm}, r) feature space the cascade uses, coloured by ground truth. **approaching** tracks (red) cluster in the positive-growth region; **moving_away** tracks (green) cluster in the negative-growth region; **crossing** tracks (blue/purple) sit at large $|dx_{norm}|$; and **uncertain** tracks (orange) cluster near the origin, where no decisive cue is available. The classifier’s dashed

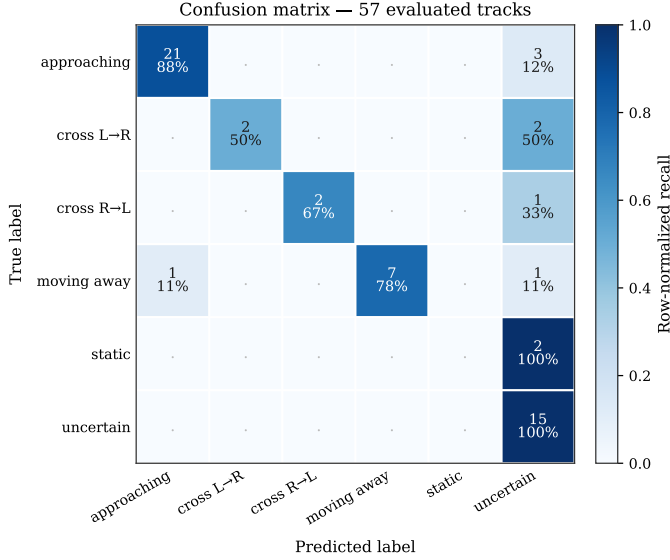


Figure 3: Confusion matrix on the 57 evaluated tracks. Off-diagonal errors are dominated by abstention into the **uncertain** column (3 **approaching**, 2 **crossing_LTR**, 1 **crossing_RTL**, 1 **moving_away**, 2 **static**). Crucially, the safety-critical **approaching**↔**moving_away** confusion is zero, and only one dangerous off-diagonal exists (**moving_away**→**approaching**, IMG_4973 / person_4).

thresholds ($|dx_{\text{norm}}| \in \{0.08, 0.50\}$ and $r \in \{-0.15, 0.15\}$) sit exactly between the class clusters—a visual confirmation that two features carry most of the decision boundary.

5.3 Ablation study

Figure 5 and Table 4 show what happens when each contribution is removed one at a time and the full pipeline is re-run. The largest improvement comes from track continuity rather than classifier complexity: appearance-gated re-ID stitching is the single largest contributor (−6.2 pts raw when removed), and the two 2-frame salvage rules each contribute ∼3.5 pts. Toggling *both* NMS and re-ID off together drops the raw number to 75.4%, a −7.0 pt joint cost, showing the two fragment-cleanup passes are complementary. The classifier-side rules (trajectory reversal, soft moving-away, slow-approach allowance) contribute ∼1.8 pts each—meaningful but not dominant. *Same-frame NMS alone barely moves the raw metric* because the overlapping-bicycle issue it addresses is rare in our data, but it is essential to keep the bookkeeping clean for downstream re-ID.

5.4 Baseline calibration and ByteTrack negative result

For calibration: random uniform across 6 classes is 16.7%, “detector-only with no motion classifier” (always **uncertain**) is 29.4%, and the majority-class baseline (always **approaching**) is 41.2%. The full system more than doubles the strongest trivial baseline.

ByteTrack on low-fps offline footage achieves only 32.8% even after tuning. Kalman-gated association assumes near-continuous detections; on 2 fps footage, predicted po-

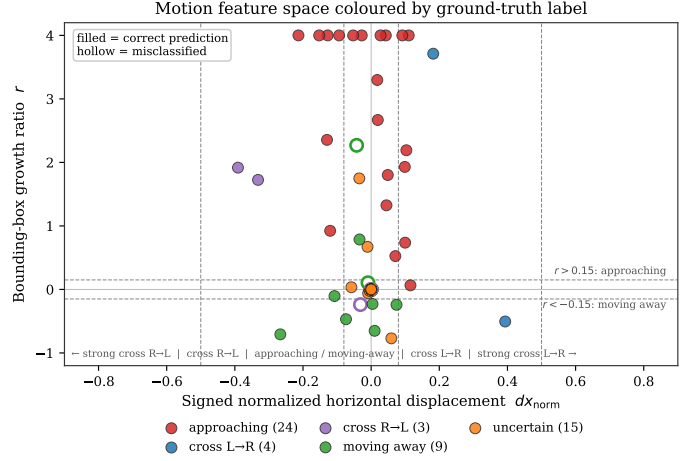


Figure 4: Motion feature space used by the rule cascade. Each point is one labelled track; filled = correct, hollow = misclassified. **approaching** occupies positive growth; **moving_away** occupies negative growth; **crossing** occupies large $|dx_{\text{norm}}|$; **uncertain** clusters near the origin. Dashed lines are classifier thresholds. Growth values are clipped to $[-1.2, 4.0]$ for readability (the four largest grow $> 4\times$).

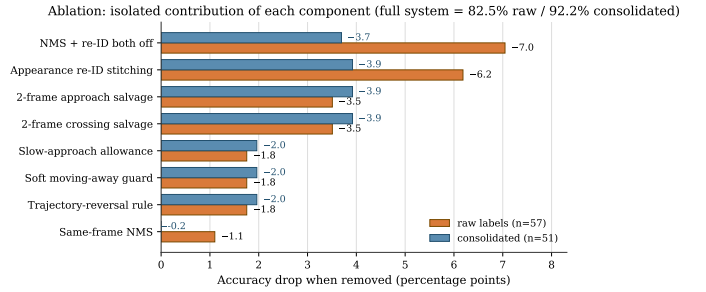


Figure 5: Ablation: accuracy drop when each component is removed from the full system, one at a time, on both the raw 57-track and consolidated 51-track denominators. Tracker-side cleanups (appearance re-ID, 2-frame salvage) drive the largest gains; classifier-side rules help less; ByteTrack underperforms on sparse low-fps footage (Section 5.4). Generated from `scripts/run_ablations.py`.

sitions over ∼500 ms gaps rarely match observed ones, and short detections never “activate” a track. ByteTrack is a useful negative result: although it is strong for many multi-object tracking settings, its assumptions are poorly matched to sparse 2 fps offline footage—in our clips it drops many low-confidence detections, reducing the temporal evidence available to the downstream classifier.

Aggressive stitching breaks established tracks (75.4% → 68.8%). An early re-ID variant allowed merging when *either* track had ≤ 4 frames; this absorbed 2-frame tracks (already correctly classified by the salvage rules) into 1-frame fragments, breaking the salvage call. Restricting to “*both* tracks must be 1 frame” kept the wins on truly fragmented fast-crossing objects without regressing any multi-frame track.

Table 4: Ablation table. Raw acc. uses the 57-track denominator; Cons. acc. uses the 51-track consolidated denominator. Macro-F1 is reported on the raw set. The largest gains come from tracker continuity (re-ID + short-track salvage); classifier rule changes contribute less.

Configuration	Change	Raw acc.	Cons. acc.	Macro F1
Full system	all components on	82.5%	92.2%	0.67
– 2-frame crossing salvage	no $n=2$ crossing rule	78.9%	88.2%	0.55
– 2-frame approach salvage	no $n=2$ approach rule	78.9%	88.2%	0.66
– Soft moving-away	tight r threshold	80.7%	90.2%	0.64
– Slow-approach allowance	no $r>0$ slow-approach	80.7%	90.2%	0.65
– Trajectory-reversal rule	no reversal heuristic	80.7%	90.2%	0.65
– Appearance re-ID	no fragment stitching	76.3%	88.2%	0.55
– Same-frame NMS	no intra-frame dedupe	81.4%	92.3%	0.65
– NMS and re-ID (both off)	fragments preserved	75.4%	88.5%	0.55



Figure 6: Qualitative examples. (a) approaching cyclist with $r = +21.98$; (b) pedestrian crossing right-to-left over 11 frames; (c) pedestrian moving away with shrinking bbox; (d) conservative abstention on a 1-frame parked-bicycle detection ($\rightarrow$ **uncertain** (**short_track**)). Yellow trails show the full centroid history; coloured boxes are the final-frame detection.

5.5 Qualitative successes and failures

Figure 6 shows one canonical example per directional class and one conservative abstention. The crossing example (b) shows the yellow centroid trail tracking sideways across more than half the frame—exactly the cue $|dx_{\text{norm}}|$ encodes. The static panel (d) is the abstention pattern that dominates the **static**-row of Figure 3: a 1-frame parked-bicycle detection that the cascade refuses to commit on.

6 Failure Analysis and Limitations

6.1 Uncertainty breakdown

Figure 7 shows the sub-reason every **uncertain** prediction carries. The distribution is heavily concentrated on **short_track** (24-frame budget across all 11 videos): 23 of the 24 **uncertain** predictions abstain because the underlying track has fewer than 3 frames, and only 1 abstains because the motion signal is too weak to commit on a longer track.

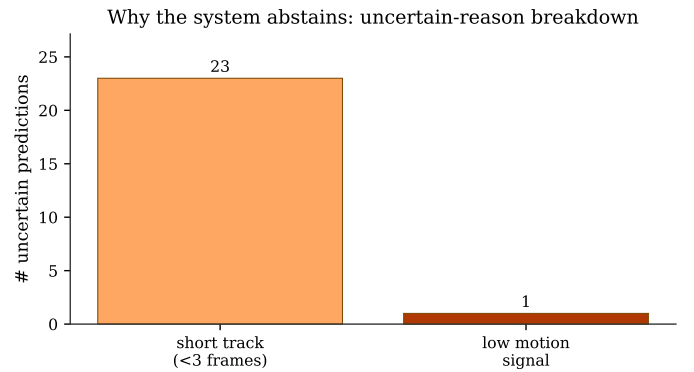


Figure 7: Sub-reason breakdown for all **uncertain** predictions. 23 of 24 abstentions are **short_track** (the track has <3 frames); only one is **low_signal**. Uncertainty in this system is dominated by tracker fragmentation, not classifier ambiguity.

This is the single most important diagnostic in the paper: uncertainty in SpatialLens is not a classifier failure mode—it is the tracker-fragmentation tax, made auditable.

6.2 Static-class result

The static-class F1 of 0.00 should not be interpreted as a general static-object failure. Only two static examples appear in the evaluation set, both as one-frame or two-frame tracker fragments (parked bicycles detected in a single frame). The cascade correctly abstains rather than committing, which is the safety-preserving behaviour we want; with the current label set it is impossible to test whether the **static** branch ever fires correctly on a long, stationary track. A larger evaluation set with multi-frame static examples is the cleanest fix.

6.3 Error taxonomy

Table 5 groups the residual errors into five root causes and lists a concrete example and proposed fix for each. Two observations matter. First, *the dominant failure is fragmentation, not misclassification*: short-track fragmentation and static abstention together account for 5 of the 10 residual errors. Second, *none of the residual errors are in the safety-critical asymmetry*: the **approaching** $\leftrightarrow$ **moving_away** confusion is zero, and the only directional mistake (**moving_away** $\rightarrow$ **approaching**, IMG_4973 / person_4) is a

monotone-bbox-growth edge case the trajectory-reversal rule misses ($\rho=1.17 < 1.5$ threshold).

6.4 Limitations

Dataset size. The evaluation set is small (61 labelled tracks, 11 videos) and class-imbalanced (only 2 `static` examples, 3 `crossing_RTL`). As a result, macro-F1 is sensitive to rare-class behaviour, and the reported numbers should be interpreted as evidence that the proposed motion features are promising—not as a deployment-level claim.

No user study. We have not yet tested whether the alert strings are useful to a blind or low-vision pedestrian under real walking conditions. The 0.95/0.88 `approaching` precision/recall is the right asymmetry for a safety alert in principle; a user-centred study is the appropriate next validation.

Low-fps regime. The pipeline is built for 2 fps offline processing. Higher sampling (5–10 fps) would alleviate most of the fragmentation residuals (Figure 7), but the threshold constants in the cascade would need to be re-tuned because they scale with inter-frame displacement.

No depth. Bounding-box growth is a noisy proxy for “getting closer.” The one residual directional mistake (`moving_away`→`approaching`) is exactly the ambiguity a monocular depth backend would resolve.

Fixed detector. We use YOLOv8n’s COCO classes unchanged. Skateboards and scooters are sometimes misclassified by the detector; the cascade inherits those errors. A small fine-tune on campus hazards would help, but it is outside the scope of an interpretable-classifier study.

7 Conclusion

SpatialLens Assist shows that interpretable motion cues can produce useful hazard labels from low-fps egocentric campus video. The system is strongest when tracks persist for at least three frames, reaching 93.3% decidable accuracy, and its primary failure mode is conservative abstention caused by tracker fragmentation rather than dangerous mislabelling: the `approaching`↔`moving_away` confusion is zero, and 23 of 24 abstentions are honest short-track aborts. The ablation study suggests that for assistive navigation at this project scale, *improving temporal track continuity matters more than replacing the interpretable classifier with a more complex learned model*—appearance-gated re-ID and short-track salvage together contribute roughly twice as much accuracy as all classifier-side rule changes combined. Future work targets the four limitations above: a monocular depth backend (MiDaS [11]) to turn bbox growth from a noisy proxy into a calibrated signal; a learned per-track embedding to replace the HSV histogram in the re-ID gate; a denser sampling regime (5–10 fps); and a user-centred evaluation of the alert layer with blind and low-vision walkers.

Individual Contributions

Solo project. Diego Sanchez authored 100% of the code, the 61-track label set, the 74-test `pytest` suite, and this

report. Specifically: all pipeline modules under `src/` (detection wrapper, IoU+centroid tracker, ByteTrack ablation backend, ego-motion compensated flow and frame-diff, per-track features with the second-half/peak-area signals, the rule-cascade classifier with evidence strings, the appearance re-ID cleanup in `src/track_reid.py`, evaluation, alerts, and visualisation); all pipeline driver scripts (`run_ablations.py` produces Figure 5 and Table 4; `make_final_figures.py` produces Figures 2 to 4, 6 and 7 and all booktabs tables); recorded the 11 controlled walkway videos and hand-labelled all 61 ground-truth tracks.

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Table 5: Error taxonomy: the five root causes that explain every residual error, with example tracks and proposed fixes. The dominant category is tracker fragmentation.

Error type	Cause	Example	Proposed fix
Short-track fragmentation	tracker links only 1–2 frames at 2 fps	IMG_4982 bicycle_1	higher fps; learned re-ID embedding
Static abstention	only 2 static examples, both 1–2 frame fragments	IMG_4982 bicycle_2	explicit stationary branch with depth cue
Crossing $\rightarrow$ uncertain	fast crossing object fragments before $ dx_{\text{norm}} $ accumulates	IMG_4977 skateboard	denser sampling; salvage at $n=1$
Moving-away $\rightarrow$ approaching	monotone bbox growth despite physical recession (no depth cue)	IMG_4973 person_4	monocular depth (MiDaS)
Trajectory-reversal miss	person approached then turned; $\rho < 1.5$	IMG_4973 person_2	softer reversal threshold; second-half flow direction